

6.4) 14) $m \in \mathbb{Z}, n \in \mathbb{Z}, m > 0$

(a) Case 1) $n \geq 0$, this follows by Theorem 6.4.1

Case 2) $n < 0$.

We can analyze this with an example. $\overset{n}{-12} = \overset{q}{2} \times \overset{m}{5} + \overset{r}{2}$. So

$-12 = -(2 \times 5 + 2) = -2 \times 5 - 2$. But we want $r = -2$ to be +ve.

We know it should be $-12 = -3 \times 5 + 3$. Note that $-r = -2$

changed to $m - r = +3$. From here, we can identify the

key is adding & subtracting m on R.H.S

$$\text{so } -n = -(+qm + r) \quad \text{By Theorem 6.4.1}$$

$$-n = -qm - r$$

$$-n = -qm - m + m - r \quad \therefore \text{Adding \& subtracting } m$$

$$-n = (-q-1)m + (m-r)$$

clearly $-q-1 \in \mathbb{Z}$ $0 \leq m-r < m$ But note that if $r=0$

then $m-r = m-0 \not< m$ so we have further 2 cases under

case 2.

Case 2.1) $r=0$ then $-n = -qm$ simply. & $r=0 < m$

Case 2.2) $r > 0$, then above reasoning applies. & $m-r < m$

(b) $n \in \mathbb{Z}, m \in \mathbb{Z}, m > 0$

$$n = qm + r \quad \wedge \quad 0 \leq r < m \quad \text{--- (1)}$$

$$n = q'm + r' \quad \wedge \quad 0 \leq r' < m \quad \text{--- (2)} \quad \rightarrow q - q' \wedge r - r'$$

There can be many ways to do this. First either (1) - (2) or putting

$n = qm + r$ in (2) gives

$$0 = (q - q')m + (r - r') \rightarrow (q - q')m = r - r' \quad \text{--- (3)}$$

We can reason by considering 2 cases ($q - q' = 0$ or $q - q' \neq 0$) or

we can use fact that $-m < 0 - r' \leq r - r' < m - r' < m$ so

$-m < r - r' < m$. This will follow easily as from (3). $m \mid r - r'$

so $\exists k \in \mathbb{Z} (km = r - r')$ so $r - r'$ is a multiple of m but as $0 < r - r' < m$
 The only multiple in between is 0. Now let's see other side of reasoning by cases.

Case 1) $q - q' \neq 0$.

From ② $\rightarrow m = \frac{r - r'}{q - q'}$ so $|m| = \left| \frac{r - r'}{q - q'} \right|$. As $q - q' \neq 0$

so $|m| \leq |r - r'|$. As $m > 0$ so $|m| = m$ so $|m| = m \leq |r - r'|$

But this contradicts the fact that $m > r - r'$ as seen above
 so this case is not possible.

Case 2) $q - q' = 0$.

Then $q = q'$ & by ② $\rightarrow 0m = r - r'$ so $r = r'$.

(c) Supp $\forall k < n$, only 1 of 3 statements is true

$k \equiv 0 \pmod{3} \rightarrow 3 | k - 0 \rightarrow \exists l (3l = k) \rightarrow k = 3l$ — ①

$k \equiv 1 \pmod{3} \rightarrow 3 | k - 1 \rightarrow 3l = k - 1 \rightarrow k = 3l + 1$ — ②

$k \equiv 2 \pmod{3} \rightarrow 3 | k - 2 \rightarrow 3l = k - 2 \rightarrow k = 3l + 2$ — ③

We have to prove this for n . Note that since $k < n$, so $n - k < n$
 so by strong induction, this is true for $n - k$, so only 1 of 3 statements is true for $n - k$. so from ①, ② & ③

① $\rightarrow n - k \rightarrow n - 3l \rightarrow n = 3l$ so $3 | n - 0$ so $n \equiv 0 \pmod{3}$

② & ③ are similar with $k = 3l + 1$ & $k = 3l + 2$

We can also prove this by dividing n by 3 so $m = 3$ & $n = 3q + r$
 then $0 \leq r < 3$ so $r = 0 \vee r = 1 \vee r = 2$ strictly. Only one of these 3 values of r is possible. In each case, either $m | n - 0$ or $m | n - 1$ or $m | n - 2$ will follow.

6.4) 9) (a) $a_0, a_1, a_2, \dots$ is Fibonacci sequence (1)

Prove: $\exists s \in \mathbb{R} \exists t \in \mathbb{R} \left[a_n = s \left(\frac{1+\sqrt{5}}{2} \right)^n + t \left(\frac{1-\sqrt{5}}{2} \right)^n \right]$ (2)

We have to find s & t from whatever information we have, that is, that $a_0, a_1, \dots$ is Fibonacci sequence.

So assuming (2) is also true, we try to find s & t using equations

we have, let $c_1 = \frac{1+\sqrt{5}}{2}$ & $c_2 = \frac{1-\sqrt{5}}{2}$ (3)

$a_0 = s c_1^0 + t c_2^0 = s + t$ (4)

$a_1 = s c_1 + t c_2$ (5)

lets solve these 2: From (4) $s = a_0 - t$. Putting this in (5)

$a_1 = c_1(a_0 - t) + t c_2 \rightarrow a_1 = a_0 c_1 - t c_1 + t c_2 \rightarrow a_1 = a_0 c_1 - t(c_1 - c_2)$

$\rightarrow t = \frac{a_0 c_1 - a_1}{c_1 - c_2}$. Putting c_1 & c_2 from (3), we get

$t = \frac{a_0 \left(\frac{1+\sqrt{5}}{2} \right) - a_1}{\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2}} \rightarrow t = \frac{2a_0 + \sqrt{5}a_0 - 2a_1}{2\sqrt{5}} \rightarrow t = \frac{a_0(1+\sqrt{5}) - 2a_1}{2\sqrt{5}}$ (6)

putting (6) in (4): $a_0 = s + \frac{a_0(1+\sqrt{5}) - 2a_1}{2\sqrt{5}} \rightarrow s = \frac{2\sqrt{5}a_0 - a_0(1+\sqrt{5}) + 2a_1}{2\sqrt{5}}$

$\rightarrow s = \frac{2\sqrt{5}a_0 - a_0 - a_0\sqrt{5} + 2a_1}{2\sqrt{5}} \rightarrow s = \frac{a_0(\sqrt{5}-1) + 2a_1}{2\sqrt{5}}$ (7)

NOW we prove (2),

Case 1) $n=0$:

$a_0 = s + t = \frac{a_0\sqrt{5} - a_0 + 2a_1}{2\sqrt{5}} + \frac{a_0 + a_0\sqrt{5} - 2a_1}{2\sqrt{5}}$

$\therefore \frac{2a_0\sqrt{5}}{2\sqrt{5}} = a_0$

Case 2) $n=1$

$a_1 = s c_1 + t c_2 = \frac{a_0\sqrt{5} - a_0 + 2a_1}{2\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right) + \frac{a_0 + a_0\sqrt{5} - 2a_1}{2\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)$

$$a_1 = \frac{a_0\sqrt{5} - a_0 + 2a_1 + 5a_0 - a_0\sqrt{5} + 2a_1\sqrt{5} + a_0 + a_0\sqrt{5} - 2a_1 - a_0\sqrt{5} - 5a_0 + 2\sqrt{5}a_1}{4\sqrt{5}}$$

$$= \frac{4a_1\sqrt{5}}{4\sqrt{5}} = a_1$$

Case 3) We will use strong induction. Supp $\forall k < n [a_k = sc_1^k + tc_2^k]$

We need to prove that $a_n = sc_1^n + tc_2^n$

$$\text{We know } a_n = a_{n-1} + a_{n-2} \quad \text{--- (8)}$$

From part (a), we know $c_1^2 = 1 + c_1$ & $c_2^2 = 1 + c_2$

$$\begin{aligned} \text{Now } sc_1^n + tc_2^n &= sc_1^{n-2}c_1^2 + tc_2^{n-2}c_2^2 \\ &= sc_1^{n-2}(1+c_1) + tc_2^{n-2}(1+c_2) \\ &= sc_1^{n-2} + sc_1^{n-1} + tc_2^{n-2} + tc_2^{n-1} \\ &= (sc_1^{n-2} + tc_2^{n-2}) + (sc_1^{n-1} + tc_2^{n-1}) \quad \therefore \text{By Inductive Hyp} \\ &= a_{n-2} + a_{n-1} \end{aligned}$$

$$sc_1^n + tc_2^n = a_n \quad \therefore \text{By (8)}$$

$$6.4) 8) (a) \quad \forall m \geq 1 \quad \forall n [F_{m+n} = F_{m-1}F_n + F_mF_{n+1}] \quad \text{--- (1)}$$

Let m be arbitrary, $m \geq 1$. We use strong induction on n

$$\text{Supp } \forall k < n [F_{m+k} = F_{m-1}F_k + F_mF_{k+1}] \quad \text{--- (2)}$$

We start by R.H.S of (1)

$$\begin{aligned} F_{m-1}F_n + F_mF_{n+1} &= F_{m-1}(F_{n-1} + F_{n-2}) + F_m(F_n + F_{n-1}) \quad \therefore \text{By def of Fibonacci Seq} \\ &= F_{m-1}F_{n-1} + F_{m-1}F_{n-2} + F_mF_n + F_mF_{n-1} \end{aligned}$$

$$= F_{m+n-1} + F_{m+n-2} \quad \therefore \text{By (2)}$$

$$= F_{m+n} \quad \therefore \text{By def of Fibonacci Seq}$$

6.4) 12) $a_0 = 0, a_1 = 1, a_2 = 1$

$$\forall n \geq 3 \left[a_n = \frac{1}{2} a_{n-3} + \frac{3}{2} a_{n-2} + \frac{1}{2} a_{n-1} \right] \quad \text{--- (1)} \quad \text{--- (2)}$$

Supp $\forall k \in \mathbb{N} \left[a_k = \frac{1}{2} a_{k-3} + \frac{3}{2} a_{k-2} + \frac{1}{2} a_{k-1} \right] \quad \text{--- (2)}$

(We start from R.H.S of (2) & apply Recursive Definitions on a_{n-3}, a_{n-2} & a_{n-1})

$$a_n = \frac{1}{2} a_{n-3} + \frac{3}{2} a_{n-2} + \frac{1}{2} a_{n-1}$$

Note that $\forall k \in \mathbb{N} \left[a_k = F_k = F_{k-1} + F_{k-2} = a_{k-1} + a_{k-2} \right] \quad \text{--- (3)}$

so $\forall k \in \mathbb{N} \left[a_k = a_{k-1} + a_{k-2} \right]$

We have to prove $a_n = F_n$

if we solve F_n & a_n separately, we will reach a point of Equality

$$F_n = F_{n-1} + F_{n-2} \quad \therefore \text{By def of } F_n$$

$$= a_{n-1} + a_{n-2} \quad \therefore \text{By (3)}$$

$$= \frac{1}{2} [a_{n-4} + 3a_{n-3} + a_{n-2}] + \frac{1}{2} [a_{n-5} + 3a_{n-4} + a_{n-3}]$$

$$= \frac{1}{2} [a_{n-2} + 2a_{n-3} + a_{n-2}] + \frac{1}{2} [a_{n-3} + 2a_{n-4} + a_{n-3}] \quad \therefore \text{By (3)}$$

$$\checkmark = \frac{1}{2} [2a_{n-1}] + \frac{1}{2} [2a_{n-2}]$$

$$= \frac{1}{2} [2a_{n-2} + 2a_{n-1}]$$

$$= \frac{1}{2} [a_{n-3} + 3a_{n-2} + a_{n-1}]$$

$$= \frac{1}{2} \left[a_{n-3} + \frac{3}{2} a_{n-2} + \frac{1}{2} a_{n-1} \right]$$

$$= a_n$$

Equality here if we solve F_n & a_n separately

6.4) 7) (a) using strong induction, supp $\forall k \in \mathbb{N} \left[\sum_{i=0}^k F_i = F_{k+2} - 1 \right] \quad \text{--- (1)}$

To prove this for n , we simply apply Recursive Def of Σ

$$\sum_{i=0}^n F_i = \sum_{i=0}^{n-1} F_i + F_n = F_{n-1+2} - 1 + F_n \quad \therefore \text{By (1) as } n-1 < n$$

$$= F_{n+1} + F_n - 1$$

$$= F_{n+2} - 1$$

$\therefore$ By Recursive Def of F_n

6.4) 13) supp given $(|P_k| = F_{k+2}) \rightarrow \textcircled{1}$

We have to prove $|P_n| = F_{n+2} \rightarrow \textcircled{2}$.

P_n is formed from $A_n = \{1, 2, 3, \dots, n\}$. There are $|P_{n-1}|$ subsets of A_n that do not contain n . All of these are in P_n . The subsets that contain n must not contain $n-1$ by definition of P_n . These can be formed by $X \cup \{n\}$ where $X \in P(A_{n-2})$ & does not contain 2 consecutive integers or in other words $X \in P_{n-2}$. So in total

$$\begin{aligned} |P_n| &= |P_{n-2}| + |P_{n-1}| = F_{n-2+2} + F_{n-1+2} && \therefore \text{By } \textcircled{1} \\ &= F_n + F_{n+1} \\ &= F_{n+2} \end{aligned}$$